NeST-Com: Fixed-Budget Next-Scale Token Communication

Supplementary Configurations and Results

S1. Configuration and calibration

This supplement provides model and protocol settings, source-paired statistics, reception accounting, generation controls, and processing costs. Image-quality results and confidence intervals retain their original source populations. The model settings, original training records, and payload summaries accompany the manuscript sources.

Models and training. NeST and WHOLE share the VAR tokenizer, VAR-d16 null-class prior, entropy coder, and adapted image decoder D_c. Its adaptation uses 20 000 ImageNet training images and continuous, fully quantized, and prefix-completion-interpolated latents. The convolutional Latent JSCC uses MSE, LPIPS, and latent-consistency losses and the same decoder; its final checkpoint is at 100k updates. SwinJSCC is trained for 200k updates on the same 20 000 images, with 256×256 input and paid side information. The selected checkpoint minimizes mean calibration MSE. Its training schedule, calibration history, checkpoint-selection records, and original training plots are available through the linked index and `inputs/training/`.

Ada adaptation. Ada retains LlamaGen-L/VQ16 and genuine autoregressive prefix arithmetic coding. A fixed ConvNeXt predicts the class at TX; its 10-bit ID travels in the protected body, giving $b + 39 \leq k_a$. RX uses the recovered class and prefix. All 1–256 independently terminated prefixes are eligible. Main-curve tail generation uses CFG=4, temperature 1, top-$k = 2000$, top-$p = 1$, FP32 with TF32 off. The fixed single-frame adaptation does not reproduce the original long-term average-budget controller or ARQ. At N1024/19 dB all 256 tokens are sent; remaining source capacity therefore does not indicate fitting failure.

NeST/WHOLE calibration. The 100-image pilot uses one channel realization per source. Final evaluation uses all 1000 calibration images with three realizations per source, including fixed-gray failures. Selection maximizes DINOv2 cosine, first averaging realizations and then equally weighting sources. Exact ties use the frozen candidate order. DINO-based selection is related to feature-based evaluation; the controlled interventions separately examine the mechanism.

N1024 target screening starts at scales 7–9 and includes full-scale $(10, 0)$ candidates in final evaluation. N512 screens all 7 WHOLE and 25 Quarter targets against 12 MCSs: $\log_2 M \in \{2, 4, 6\}$ and rates $\{1/2, 2/3, 3/4, 5/6\}$. Each method/SNR advances the top three and a maximum-endpoint QPSK rate-1/2 anchor. Ada uses the same N512 MCS pool and up to 256 tokens. Each budget is independently calibrated; its results describe separately selected systems, rather than an isolated causal change in budget or representation order.

Swin checkpoint and training status; training and calibration curves; N512 frozen protocol.

S1.1. Frozen model, coder and policies

The following configuration is verified against the actual NeST/WHOLE loader and formal receiver receipts. State SHA-256 prefixes identify the weights below; full hashes, file hashes, numerical flags and exact implementation paths are supplied in `results/nestcom_focused_review_20261029_v2/configuration_audit.json`.

Component	Frozen setting
Prior	VAR-d16; null class; greedy completion without CFG
Tokenizer	4096 entries; 32 latent channels; ten scales
Numerical route	FP32 model logits; TF32 off; deterministic kernels; flash/fused attention off
CDF	CPU float64 softmax; total 2^{24}; minimum frequency one
Integer frequencies	$\lfloor p_i(2^{24} - 4096) \rfloor + 1$; all remaining mass goes to the largest-probability token (first index for a tie)
Arithmetic coder	32-bit interval; independent termination per candidate
VAE state	`6830fc533a34d52c`
VAR state	`b82cb0855b24e1d0`
D_c state	`bf1d64bf8ff7eeda`
Runtime	NeST/WHOLE receipt: Python 3.10.12, PyTorch 2.11.0+cu128, CUDA 12.8

The entropy-decoding CDF does not use generation-stage CFG. These identities specify the numerical implementation; FP32 alone is not a guarantee of cross-hardware bitwise agreement.

N	Method	SNR	Target	$\log_2 M$	Rate
512	Ada	4	256 tokens	2	2/3
512	NeST	4	$(5, 27)$	2	2/3
512	WHOLE	4	$(6, 0)$	2	2/3
512	Ada	10	256 tokens	4	2/3
512	NeST	10	$(7, 0)$	4	2/3
512	WHOLE	10	$(7, 0)$	4	2/3
512	Ada	19	256 tokens	6	5/6
512	NeST	19	$(8, 84)$	6	5/6
512	WHOLE	19	$(8, 0)$	6	5/6
1024	Ada	4	256 tokens	2	2/3
1024	NeST	4	$(7, 25)$	2	2/3
1024	WHOLE	4	$(7, 0)$	2	2/3
1024	Ada	10	256 tokens	4	2/3
1024	NeST	10	$(8, 126)$	4	2/3
1024	WHOLE	10	$(9, 0)$	4	2/3
1024	Ada	19	256 tokens	6	5/6
1024	NeST	19	$(10, 0)$	6	5/6
1024	WHOLE	19	$(10, 0)$	6	5/6

These are calibrated targets. Online fitting can select a shorter legal endpoint under the same MCS. SNR is in dB. Full candidate identifiers are retained in `frozen_policies_compact.csv`.

S1.2. Six-system adaptations and selection

The main comparison evaluates complete systems with different pretraining and adaptation. The tables distinguish these resources from the shared-model WHOLE/NeST comparison. No configuration is changed by this revision.

System	Representation and adaptation
NeST / WHOLE	VAR-d16 and residual VQ tokenizer; shared adapted D_c. D_c: 20,000 ImageNet training images; continuous/full-quantized/prefix-completed interpolated latents; all components frozen at evaluation
Latent JSCC	Frozen VAR tokenizer continuous 32 x 16 x 16 latent; convolutional channel encoder/decoder; shared D_c. 495,208 trainable communication parameters; width 64, 3 residual blocks; 20,000 training images; final 100,000 updates; MSE + 0.1 LPIPS + 0.01 standardized-latent MSE
SwinJSCC	Author SA+RA architecture adapted to 256 x 256, initialized without pretrained communication weights. 20,000 training images; C6 / C13 alternating budgets; 200,000 total updates; RGB MSE; checkpoint selected by minimum calibration MSE
BPG + LDPC	libbpg 0.9.8 / x265; 8-bit YCbCr 4:2:0, compression level 8. No neural training; image-adaptive downsampling to 256 / 128 / 64 / 32 and QP search 0–51
ARPC adapted	Official ARPC adapted Infinity 2B/BSQ-VAE 16 weights; BLIP caption and Flan-T5-XL text conditioning. Frozen public communication weights; fixed-frame container and actual LDPC/QAM interface; no new training
Ada adapted	LlamaGen-L / VQ16; fixed ConvNeXt predicts ImageNet class. Frozen pretrained representation and prior; fixed per-image frame, actual prefix arithmetic coding and paid class; no ARQ/controller reproduction

SwinJSCC is trained for 200,000 updates and selected using the mean RGB MSE over 1,000 calibration images, two symbol budgets, and six SNRs. Its late-stage training loss remains near 0.0061, and mean calibration MSE decreases by 0.20% over the final 20,000 updates. The selected checkpoint, training curves, and stopping records remain linked in S1; the accompanying `inputs/training/` directory provides the original numerical records and the aggregation audit. The decoder adaptation for NeST/WHOLE does not define a separately trained coarse-only branch.

S1.2. Transmission and calibration (continued)

System	Candidates and channel mapping	Selection and receiver
NeST / WHOLE	N512: 25 Quarter / 7 WHOLE; N1024: family-specific targets and full-scale m = 10 candidates, S1.1. Discrete LDPC/QAM; N512 12 MCSs; N1024 frozen pool and targets in S1.1	DINOv2 cosine; cal1000, three channel outcomes/source; 100-source pilot one outcome. Profile endpoint/MCS; body 13-bit length +16-bit CRC. Null class, greedy next-scale completion; received-token overwrite and shared D_c
Latent JSCC	No discrete token-prefix candidates. Continuous 1024 complex symbols, per-image Es = 2 normalization; no MCS	Checkpoint calibration on 1000 images, 5 training-SNR cells, 3 noises; frozen composite loss. No predicted class or text. Learned latent recovery and frozen D_c; no autoregressive tail
SwinJSCC	No discrete token-prefix candidates. N1024: 768 continuous data symbols plus 256 charged mask/power side-information symbols; no MCS	1000 calibration images across 2 budgets and 6 SNRs, one noise per cell; mean RGB MSE. Mask and power side information charged within N. Learned image decoder; no token completion
BPG + LDPC	Coarse QP 0, 8, 16, 24, 32, 40, 48, 51 plus integer refinement at capacity crossings; select legal minimum-MSE stream. 15 MCS catalogue: QPSK/16-/64-QAM x rates 1/3, 1/2, 2/3, 3/4, 5/6; three screened per SNR	100 calibration images, 3 noises; maximum source-mean PSNR; initial 32-source proxy screen. Complete BPG file plus protected framing; no oracle RX size or source. Actual BPG decoding and bicubic resize to 256 x 256; rejected receptions gray
ARPC adapted	7 complete-scale endpoints; one probability pass; 16-bit group trial arithmetic coding vs raw. 12 MCSs: QPSK/16-/64-QAM x rates 1/2, 2/3, 3/4, 5/6; 2 shortlisted/SNR	100-source, 84-strategy proxy screen, then cal1000 x 3 actual noises; maximum DINOv2 cosine. Caption and token boundaries included in lossless source container. Actual prefix plus paid caption; CFG 3, temperature 0.5, top-k 900, top-p 0.97, seed 0; full scale schedule
Ada adapted	All 1–256 independently terminated raster prefixes. N1024: 6 preselected MCS/SNR combinations; N512: 12 MCS pool in S1.1	cal1000 x 3 actual noises; maximum DINOv2 cosine, then freeze. Recovered 10-bit predicted class; body overhead 39 bits including length/CRC. CFG 4, temperature 1, top-k 2000, top-p 1 sampled tail; FP32 / TF32 off

BPG's final 100-source PSNR calibration differs from the 1000-source DINO calibration of the generative systems. Continuous JSCC systems do not have a token-prefix/MCS search. None of these differences is treated as an isolated representation-order effect. Full settings and source SHA-256 identities are in `six_system_configuration.csv` and `configuration_evidence.json` in the V3 result directory.

S1.3. ARPC caption and visual payload accounting

This descriptive audit uses the existing `external_common100` sources contained in the common500 study. For each source and SNR, the recorded endpoint is joined to the original source-container breakdown. The 900 source/SNR/repetition identities agree with the common500 reception records, and the 300 source/SNR container lengths agree with the source encoder records. Repetitions of an identical source payload are counted once when computing the equal-source means.

The encoded caption content averages 258.64 bits, with a range of 96–424 bits. A one-bit coding-mode flag and nine-bit length field bring the mean caption contribution to 268.64 bits. The table reports the source container before the outer 29-bit length/CRC overhead, LDPC, and protected physical profile. Percentages are the means of per-source component/container ratios, rather than ratios of the displayed means. The remaining source-container bits contain its other recorded fields.

SNR (dB)	Container (bit)	Visual data (bit)	Caption share	Visual share
4	950.48	571.26	31.64%	56.80%
10	2245.36	1672.79	11.84%	74.57%
19	4516.88	3808.86	6.12%	84.10%

These values describe the fixed-frame operating points of the 100-source audit. The main image-quality comparison continues to use all 500 sources. Exact component lengths, joins, and source hashes are provided in `inputs/review_audit/arpc_common100_actual_payload_breakdown.csv`, `inputs/review_audit/arpc_common100_actual_payload_summary.csv`, and the accompanying receipt.

S2. Statistical populations and reception accounting

The main population is common500; Kodak has 24 sources. The endpoint, receiver, and E1/E2 controls use the existing fixed 100-source subset, not a new independent blind test. Full source IDs, preprocessing identities, and native pixel hashes join the records. Three original outcomes are averaged within a source before equal-source means and paired differences. Percentile 95% intervals use 10 000 source resamples; they are pointwise, not simultaneous intervals. Marginal intervals are never subtracted.

N512 methods share the registered source/repeat AWGN identities. N1024 preserves historical noise identities and is not presented as sharing noise across every method or budget. E1/E2 reuse those actual reception states; receiver counterfactuals are not new wireless frames. The 100-source bank is matched by its complete source order, not sliced from the 500-source bank. Both registered banks use seed 2026100701. Full hashes and source orders remain in the statistical identity files.

Main matched-SNR failures. N1024 has no rejected or failed reconstructions in 1500 trials per method/SNR; continuous Latent JSCC has no discrete CRC rejection. N512 failures below are body-CRC rejections, retained as fixed gray in all means. Header, parser, and source-decoder rejection counts are zero. No evaluated N512 source requires a NULL frame.

Method	4 dB	10 dB	19 dB	Trials per SNR
NeST	4	1	0	1500
WHOLE	7	0	0	1500
Ada	5	0	0	1500

Historical no-feasible-endpoint behavior. The evaluated implementations differ. N512 sends a paid NULL profile and two zero payload bits through the complete 68+444-symbol frame, returning gray. The frozen N1024 implementation raises an execution error when no independently terminated legal stream fits; no evaluated N1024 source encountered this case. The historical implementations and results remain unchanged. The separately versioned N1024 protocol extension and its bounded checks are documented in S7.

Granularity and receiver controls. Mean terminated-length checks are 3.55/5.55/9.49/17.58 for Whole/Half/Quarter/Eighth. They compare already terminated streams, not extra model forwards or measured latency. Quarter is a predefined configuration, not a demonstrated optimum. The A0/B0/C0 differences are ordered interventions on the same prefixes; they are not a universal percentage decomposition. A0 still performs prior completion. At the final scale C0 and B0 coincide because no finer context is used.

S2.1. Actual formal-frame energy

The transmitted body follows packing, LDPC/RV0 rate matching, public mask XOR, and QAM modulation. The receiver reverses the mask by changing the LLR signs before LDPC decoding. The deterministic mask uses public protocol/session/frame-counter identities; it is not secret side information. Its implementation identifier is `WCL_T1_public_PCG64_SHA256_protocol_session_counter_after_RV0_v1`.

The following table joins the current formal source/policy/repetition records to their original physical-event files and stored energies. It covers 22 500 logical frames and 20 613 distinct physical-event paths: N1024 WHOLE/PARTIAL and N512 WHOLE/NeST/Ada at 4/10/19 dB. Each row equally weights 500 sources with three repetitions each. Every method/SNR row has 1500 distinct event paths; some paths are shared between arms, explaining the smaller cross-arm total. Method-group logical and physical-unique summaries are both provided in the CSV. No older-policy energy table substitutes for these events.

Let $\rho = E_{\text{frame}}/(2N)$. The table gives its mean, population standard deviation (SD, divisor 1500), minimum and maximum, together with mean E_{frame}/N. The corresponding SD and extremes of E_{frame}/N are twice those of ρ; all values are retained in the CSV.

N	Method	SNR	Mean E/N	Mean ρ	SD ρ	Min ρ	Max ρ
512	Ada	4	2.000000	1.000000	0.000000	1.000000	1.000000
512	Ada	10	2.000208	1.000104	0.022871	0.939062	1.089062
512	Ada	19	1.998569	0.999285	0.026561	0.918155	1.087798
512	NeST	4	2.000000	1.000000	0.000000	1.000000	1.000000
512	NeST	10	1.999427	0.999713	0.023503	0.920312	1.068750
512	NeST	19	2.000920	1.000460	0.025730	0.909970	1.088170
512	WHOLE	4	2.000000	1.000000	0.000000	1.000000	1.000000
512	WHOLE	10	2.002316	1.001158	0.023419	0.928125	1.081250
512	WHOLE	19	2.001408	1.000704	0.025361	0.925223	1.086681
1024	PARTIAL	4	2.000000	1.000000	0.000000	1.000000	1.000000
1024	PARTIAL	10	2.001052	1.000526	0.017186	0.945312	1.057031
1024	PARTIAL	19	2.000258	1.000129	0.018466	0.930617	1.063058
1024	WHOLE	4	2.000000	1.000000	0.000000	1.000000	1.000000
1024	WHOLE	10	2.000459	1.000230	0.017047	0.945312	1.057031
1024	WHOLE	19	1.999787	0.999894	0.018440	0.930617	1.063058

All 4500 N512/4 dB logical frames use QPSK and have exactly $E_{\text{frame}} = 1024$ and $\rho = 1$; the 68 header symbols each have energy two. Higher-order QAM exhibits finite-frame energy variation. The largest absolute paired-arm ratio of mean energies below is 0.0063 dB; the means are close without asserting equality of individual frame energies.

N	SNR	$10\log_{10}(\bar{E}_P/\bar{E}_W)$ (dB)	Mean $\Delta\rho$	$r(\Delta E, \Delta p)$
512	4	+0.000000	+0.000000	—
512	10	-0.006272	-0.001445	+0.0332
512	19	-0.001060	-0.000244	+0.0335
1024	4	+0.000000	+0.000000	—
1024	10	+0.001287	+0.000296	-0.0144
1024	19	+0.001023	+0.000235	-0.0259

Here P is PARTIAL/NeST, W is WHOLE, and p is the zero-padding fraction of body information bits. The paired quantities use the same source/repetition identities; r is a descriptive Pearson correlation across the 1500 pairs. A dash denotes zero energy variance. Across the nonconstant method/SNR groups, the absolute correlation between ρ and padding fraction is at most 0.042. These summaries characterize the recorded transmissions; no per-frame normalization or new power-matched experiment was introduced.

S2.2. Actual endpoints and retained source content

At N512/10 dB, WHOLE and NeST share target $(7, 0)$ and 16-QAM rate 2/3. Direct checks of all three repetitions confirm that every source has the same TX endpoint across its repeats. The following counts are therefore source counts, not independent frame observations.

Actual endpoint	WHOLE sources	NeST sources
$(6, 0)$	498	53
$(6, 16)$	0	378
$(6, 32)$	0	59
$(6, 48)$	0	8
$(7, 0)$	2	2
Total	500	500

Both methods fall back from the target on 498/500 sources (99.6%). NeST retains a within-scale endpoint on 445/500 (89.0%). The first six scales contain 91 tokens and the seventh contains 64. Mean retained-token counts are 91.256 for WHOLE and 107.896 for NeST, a difference of 16.640. Source-bit utilization rises from 79.30% to 93.46%. The finer fallback retains additional information even when a full next scale does not fit; it does not reduce the target-fallback rate at this point.

The N1024 matched-SNR gain occurs with zero rejections in both families at every tested SNR. At N512/19 dB, both have zero rejections and NeST still gains 0.792 dB. At N512/10 dB, NeST has one additional body-CRC rejection yet gains 0.399 dB in the unconditional mean. These observations retain all fixed-gray failures and do not replace the primary population with a success-only subset.

Full-precision energies, paired energy differences, padding correlations, failure counts, source/endpoint counts, original-file hashes and the logical-to-physical mapping are in `results/nestcom_focused_review_20261029_v2/`.

S2.3. Quality decomposition by reception status

These calculations use only the existing N512 native-array metric scores; no decoder, metric model, channel, or bootstrap is run. Every method/SNR has 500 sources and three repetitions. With frame-equal success and failure means,

$$Q_{\text{all}} = p_{\text{success}} Q_{\text{success}} + (1 - p_{\text{success}}) Q_{\text{failure}}.$$

Because each source has three outcomes, the original equal-source overall mean equals this frame-equal mean. The maximum numerical residual of the identity over all 36 cells is 1.12×10^{-16}. Gray-image scores are taken from each actual failed event; they need not be equal across sources. A dash means no failed frame, with zero weighted failure term. **LPIPS-Alex**

Method	SNR	Success/1500	Overall	Success frames	Failed frames
ADA	4	1495	0.554156	0.552943	0.916893
ADA	10	1500	0.410977	0.410977	–
ADA	19	1500	0.214568	0.214568	–
NEST	4	1496	0.365175	0.363679	0.924583
NEST	10	1499	0.272596	0.272076	1.052495
NEST	19	1500	0.190053	0.190053	–
WHOLE	4	1493	0.395124	0.392485	0.958144
WHOLE	10	1500	0.288079	0.288079	–
WHOLE	19	1500	0.217542	0.217542	–

DreamSim

Method	SNR	Success/1500	Overall	Success frames	Failed frames
ADA	4	1495	0.384728	0.382844	0.948163
ADA	10	1500	0.258171	0.258171	–
ADA	19	1500	0.107753	0.107753	–
NEST	4	1496	0.281865	0.280131	0.930711
NEST	10	1499	0.192165	0.191687	0.909004
NEST	19	1500	0.121676	0.121676	–
WHOLE	4	1493	0.324663	0.321794	0.936442
WHOLE	10	1500	0.204857	0.204857	–
WHOLE	19	1500	0.145797	0.145797	–

S2.3. Quality decomposition (continued)

PSNR (dB)

Method	SNR	Success/1500	Overall	Success frames	Failed frames
ADA	4	1495	11.318311	11.314918	12.332928
ADA	10	1500	13.164679	13.164679	–
ADA	19	1500	17.690030	17.690030	–
NEST	4	1496	15.498715	15.507571	12.186599
NEST	10	1499	17.219878	17.219739	17.428954
NEST	19	1500	19.429779	19.429779	–
WHOLE	4	1493	15.106545	15.120288	12.175318
WHOLE	10	1500	16.821305	16.821305	–
WHOLE	19	1500	18.637774	18.637774	–

CLIP-I

Method	SNR	Success/1500	Overall	Success frames	Failed frames
ADA	4	1495	0.789210	0.790088	0.526638
ADA	10	1500	0.825748	0.825748	–
ADA	19	1500	0.872282	0.872282	–
NEST	4	1496	0.765689	0.766235	0.561732
NEST	10	1499	0.814979	0.815175	0.521134
NEST	19	1500	0.864000	0.864000	–
WHOLE	4	1493	0.746943	0.748068	0.507109
WHOLE	10	1500	0.807443	0.807443	–
WHOLE	19	1500	0.848354	0.848354	–

S2.3. Joint-success source-paired comparisons

For each comparison, events are joined by complete source ID and original repetition identity. Only events where both methods succeed are retained. Differences are averaged over the available joint-success repetitions within each source, then equally over sources. These are descriptive conditional comparisons with no new confidence interval; they do not replace the failure-inclusive primary estimates.

Comparison	SNR	Joint frames	Sources	Two repeats	Three repeats
NEST-WHOLE	4	1493	500	7	493
NEST-WHOLE	10	1499	500	1	499
NEST-WHOLE	19	1500	500	0	500
NEST-ADA	4	1491	500	9	491
NEST-ADA	10	1499	500	1	499
NEST-ADA	19	1500	500	0	500

Every source retains at least two jointly successful repeats; no source is discarded.

Comparison	SNR	ΔLPIPS	ΔDreamSim	ΔPSNR	ΔCLIP-I
NEST-WHOLE	4	-0.02894	-0.04174	+0.38886	+0.01825
NEST-WHOLE	10	-0.01602	-0.01315	+0.39936	+0.00773
NEST-WHOLE	19	-0.02749	-0.02412	+0.79200	+0.01565
NEST-ADA	4	-0.18923	-0.10249	+4.19030	-0.02398
NEST-ADA	10	-0.13891	-0.06646	+4.05598	-0.01057
NEST-ADA	19	-0.02451	+0.01392	+1.73975	-0.00828

The LPIPS and DreamSim directions agree with the original all-event contrasts at all reported cells. Thus the observed quality differences persist on joint successes; the comparison still includes the differing models and training of each complete system. Files: `success_failure_quality.csv`, `joint_success_paired_quality.csv`, and the source-level differences and identity JSON in the V3 `reception/` directory.

S2.4. Perceptual endpoint and receiver evidence

The following are existing source-paired bootstrap results, reorganized for the perceptual-reconstruction narrative. No statistic is recomputed. All common500 LPIPS and DreamSim intervals favor NeST/within-scale transmission at both budgets and all three SNRs. This claim does not extend to every dataset or metric; the N1024 common500 pair file does not provide CLIP-I.

N	SNR	ΔLPIPS [95% CI]	ΔDreamSim [95% CI]
1024	4	-0.02458 [-0.02621,-0.02300]	-0.02073 [-0.02328,-0.01827]
1024	10	-0.02725 [-0.02962,-0.02491]	-0.02405 [-0.02700,-0.02124]
1024	19	-0.00520 [-0.00563,-0.00478]	-0.00304 [-0.00372,-0.00242]
512	4	-0.02995 [-0.03366,-0.02640]	-0.04280 [-0.04830,-0.03734]
512	10	-0.01548 [-0.01703,-0.01374]	-0.01269 [-0.01496,-0.01041]
512	19	-0.02749 [-0.02883,-0.02616]	-0.02412 [-0.02639,-0.02196]

Differences are PARTIAL/NeST minus WHOLE; populations are 500 sources, each with three original channel outcomes. The 13.7% LPIPS and 18.3% DreamSim reductions at N1024/10 dB describe relative changes of metric means, not percentages of human perceptual improvement.

Controlled granularity at N1024/10 dB, 100 sources.

Grid	Util. (%)	LPIPS	DreamSim	PSNR	Length checks
Whole	78.15	0.201821	0.131529	18.951576	3.55
Half	89.21	0.183760	0.114775	19.395750	5.55
Quarter	95.38	0.175170	0.107702	19.611725	9.49
Eighth	97.39	0.172939	0.106972	19.734671	17.58

Eighth–Quarter differences are LPIPS $-0.002231\,[-0.003228, -0.001310]$, DreamSim -0.000730 $[-0.001956, 0.000477]$, and PSNR $+0.122946\,[0.078557, 0.169317]$ dB. A DreamSim interval crossing zero establishes neither equivalence nor saturation. Quarter remains the predefined main configuration.

Ordered receiver intervention at N1024/10 dB.

Intervention	ΔLPIPS [95% CI]	ΔDreamSim [95% CI]
B0–A0	$-0.010030\,[-0.012026, -0.008026]$	$-0.008454\,[-0.011357, -0.005643]$
C0–B0	$-0.016620\,[-0.020494, -0.012862]$	$-0.015372\,[-0.019880, -0.011113]$

The 100-source intervention order is fixed; these are not order-independent causal percentages. The final-scale C0–B0 effect at 19 dB is zero.

S2.5. Transfer to Kodak24

ImageNet-calibrated policies are applied unchanged to the 24 Kodak sources at $N = 1024$. Differences are PARTIAL minus WHOLE. Each source is averaged over three channel outcomes before the direct source-paired 95% confidence interval is computed. Utilization is the source-bit capacity fraction for WHOLE (W) and PARTIAL (P).

SNR (dB)	Util. W→P (%)	ΔLPIPS [95% CI]
4	75.69 → 93.43	−0.0189 [−0.0247, −0.0130]
10	78.11 → 94.99	−0.0267 [−0.0370, −0.0167]
19	85.69 → 93.71	−0.0038 [−0.0056, −0.0022]

S3. Ada tail sensitivity: $N = 512$, 4 dB

All entries use the same 100 sources and three original repetitions (300 logical results per arm). Each cell is a mean or direct paired difference with its 95% source-bootstrap interval. Higher PSNR/CLIP-I and lower LPIPS/DreamSim are favorable.

Absolute metrics

Metric	Ada sampling	Ada argmax	NeST
PSNR (dB) ↑	10.984 [10.574, 11.410]	10.509 [10.099, 10.933]	15.141 [14.621, 15.673]
LPIPS ↓	0.5610 [0.5457, 0.5759]	0.6033 [0.5857, 0.6209]	0.3668 [0.3511, 0.3823]
DreamSim ↓	0.3797 [0.3523, 0.4071]	0.4247 [0.3966, 0.4534]	0.2786 [0.2591, 0.2981]
CLIP-I ↑	0.7922 [0.7740, 0.8095]	0.7661 [0.7466, 0.7851]	0.7658 [0.7468, 0.7855]

Direct source-paired differences

Metric	Argmax minus sampling	NeST minus argmax	NeST minus sampling
PSNR (dB) ↑	-0.475 [-0.801, -0.152]	4.632 [4.183, 5.093]	4.157 [3.772, 4.558]
LPIPS ↓	0.0423 [0.0294, 0.0555]	-0.2365 [-0.2567, -0.2167]	-0.1942 [-0.2100, -0.1789]
DreamSim ↓	0.0450 [0.0282, 0.0621]	-0.1461 [-0.1709, -0.1218]	-0.1010 [-0.1248, -0.0769]
CLIP-I ↑	-0.0261 [-0.0378, -0.0143]	-0.0003 [-0.0194, 0.0185]	-0.0263 [-0.0449, -0.0078]

Fixed-gray counts per 300: Ada-sampling: 0, Ada-argmax: 0, NeST: 1. The deterministic tail lowers Ada PSNR and worsens its three feature metrics relative to sampling. NeST retains higher PSNR and lower LPIPS/DreamSim than argmax; the CLIP-I difference interval crosses zero. The sampling configuration remains the main baseline. Only unreceived-tail token selection changes. Context/KV is rebuilt from the actual received prefix and follows each new greedy token. The paid class, CFG composition, entropy-decoding conditional CDF, model, decoder, and numerical route remain fixed. No sampling-tail KV is reused. Full-token receptions reuse the existing output exactly; failed/NULL inputs remain gray.

S3. Ada tail sensitivity: $N = 512$, 10 dB

All entries use the same 100 sources and three original repetitions (300 logical results per arm). Each cell is a mean or direct paired difference with its 95% source-bootstrap interval. Higher PSNR/CLIP-I and lower LPIPS/DreamSim are favorable.

Absolute metrics

Metric	Ada sampling	Ada argmax	NeST
PSNR (dB) ↑	12.878 [12.384, 13.398]	12.227 [11.738, 12.751]	16.827 [16.313, 17.347]
LPIPS ↓	0.4121 [0.3975, 0.4266]	0.4416 [0.4237, 0.4598]	0.2818 [0.2683, 0.2955]
DreamSim ↓	0.2511 [0.2296, 0.2728]	0.2856 [0.2627, 0.3098]	0.1955 [0.1797, 0.2123]
CLIP-I ↑	0.8337 [0.8170, 0.8502]	0.8105 [0.7929, 0.8278]	0.8208 [0.8028, 0.8386]

Direct source-paired differences

Metric	Argmax minus sampling	NeST minus argmax	NeST minus sampling
PSNR (dB) ↑	-0.651 [-0.984, -0.338]	4.600 [4.188, 5.018]	3.949 [3.566, 4.348]
LPIPS ↓	0.0296 [0.0201, 0.0394]	-0.1598 [-0.1754, -0.1439]	-0.1302 [-0.1429, -0.1178]
DreamSim ↓	0.0345 [0.0232, 0.0461]	-0.0901 [-0.1120, -0.0689]	-0.0556 [-0.0760, -0.0360]
CLIP-I ↑	-0.0231 [-0.0337, -0.0128]	0.0102 [-0.0072, 0.0280]	-0.0129 [-0.0300, 0.0039]

Fixed-gray counts per 300: Ada-sampling: 0, Ada-argmax: 0, NeST: 1. The deterministic tail lowers Ada PSNR and worsens its three feature metrics relative to sampling. NeST retains higher PSNR and lower LPIPS/DreamSim than argmax; the CLIP-I difference interval crosses zero. The sampling configuration remains the main baseline. Only unreceived-tail token selection changes. Context/KV is rebuilt from the actual received prefix and follows each new greedy token. The paid class, CFG composition, entropy-decoding conditional CDF, model, decoder, and numerical route remain fixed. No sampling-tail KV is reused. Full-token receptions reuse the existing output exactly; failed/NULL inputs remain gray.

S3. Ada tail sensitivity: $N = 1024$, 10 dB

All entries use the same 100 sources and three original repetitions (300 logical results per arm). Each cell is a mean or direct paired difference with its 95% source-bootstrap interval. Higher PSNR/CLIP-I and lower LPIPS/DreamSim are favorable.

Absolute metrics

Metric	Ada sampling	Ada argmax	NeST
PSNR (dB) ↑	19.200 [18.592, 19.824]	19.044 [18.403, 19.698]	19.612 [19.083, 20.146]
LPIPS ↓	0.1717 [0.1596, 0.1845]	0.1755 [0.1625, 0.1890]	0.1752 [0.1653, 0.1851]
DreamSim ↓	0.0863 [0.0795, 0.0933]	0.0899 [0.0823, 0.0978]	0.1077 [0.1000, 0.1157]
CLIP-I ↑	0.8897 [0.8764, 0.9023]	0.8884 [0.8751, 0.9008]	0.8759 [0.8607, 0.8905]

Direct source-paired differences

Metric	Argmax minus sampling	NeST minus argmax	NeST minus sampling
PSNR (dB) ↑	-0.156 [-0.369, 0.057]	0.568 [0.246, 0.902]	0.412 [0.157, 0.677]
LPIPS ↓	0.0038 [0.0007, 0.0071]	-0.0003 [-0.0072, 0.0062]	0.0034 [-0.0021, 0.0089]
DreamSim ↓	0.0036 [0.0010, 0.0060]	0.0178 [0.0129, 0.0229]	0.0214 [0.0167, 0.0262]
CLIP-I ↑	-0.0013 [-0.0052, 0.0025]	-0.0124 [-0.0222, -0.0025]	-0.0137 [-0.0235, -0.0038]

Fixed-gray counts per 300: Ada-sampling: 0, Ada-argmax: 0, NeST: 0. The argmax PSNR difference from sampling includes zero, while DreamSim becomes worse. NeST retains higher PSNR than argmax; Ada argmax retains better DreamSim and CLIP-I, and the LPIPS interval crosses zero. Decoding sensitivity does not remove other complete-system differences. Only unreceived-tail token selection changes. Context/KV is rebuilt from the actual received prefix and follows each new greedy token. The paid class, CFG composition, entropy-decoding conditional CDF, model, decoder, and numerical route remain fixed. No sampling-tail KV is reused. Full-token receptions reuse the existing output exactly; failed/NULL inputs remain gray.

S4. Completion versus direct coarse decoding

R0 decodes $D_c(\sum_{s=1}^{m} C_s(q_s))$ from the actual received WHOLE prefix. R1 is the existing VAR-completed reconstruction of that same prefix. Both retain the original ten-scale indexing, scale-dependent residual transforms, full latent grid, frozen adapted D_c, clipping, and metric preprocessing. Unreceived scales contribute nothing to R0; codebook ID zero is not used as zero contribution. Neither arm uses unsent source tokens or a continuous source latent. This is distinct from A0/B0/C0, which intervene on received boundary-token use.

Decoder training coverage. The fixed decoder was adapted with an exact 50% continuous-latent, 25% fully quantized-latent, and 25% interpolation mixture. The interpolation starts from $F_{b,\mathrm{TX}}$, a complete latent formed by `complete_latent` from the first eight true source scales and generated subsequent scales, and moves toward the continuous latent. That training-cache completion uses the training source's class label, whereas the formal NeST receiver uses null-class greedy completion without CFG. The interpolation does not start from a coarse-only residual sum. The training code has no explicit coarse-only input branch. Sharing D_c therefore fixes the parameters but does not equalize adaptation to the two input distributions.

$N = 512$, 4 dB: 100 sources, 300 results per arm

Metric	R0 coarse	R1 completed	R1 minus R0
PSNR (dB) ↑	14.787 [14.293, 15.296]	14.813 [14.261, 15.383]	0.026 [-0.204, 0.271]
LPIPS ↓	0.6612 [0.6376, 0.6843]	0.3960 [0.3752, 0.4172]	-0.2652 [-0.2806, -0.2500]
DreamSim ↓	0.7369 [0.7154, 0.7575]	0.3166 [0.2915, 0.3421]	-0.4204 [-0.4488, -0.3925]
CLIP-I ↑	0.5493 [0.5337, 0.5649]	0.7521 [0.7303, 0.7737]	0.2028 [0.1837, 0.2218]

Fixed-gray count per arm: 3.

$N = 512$, 10 dB: 100 sources, 300 results per arm

Metric	R0 coarse	R1 completed	R1 minus R0
PSNR (dB) ↑	15.722 [15.235, 16.209]	16.493 [15.993, 17.004]	0.771 [0.563, 0.986]
LPIPS ↓	0.5581 [0.5388, 0.5771]	0.2943 [0.2817, 0.3068]	-0.2638 [-0.2774, -0.2504]
DreamSim ↓	0.6574 [0.6335, 0.6804]	0.2055 [0.1897, 0.2216]	-0.4519 [-0.4790, -0.4248]
CLIP-I ↑	0.5748 [0.5574, 0.5924]	0.8150 [0.7965, 0.8338]	0.2403 [0.2206, 0.2595]

Fixed-gray count per arm: 0. Within this fixed adapted receiver, completion improves all three feature metrics at both points; PSNR improves clearly at 10 dB, while its 4 dB interval includes zero. This controlled switch does not establish superiority over a coarse-only decoder trained with comparable input coverage. No new decoder training was performed for this revision.

S5. Additional results and processing cost

N2048. On 100 sources, the existing experiment uses two paid 1024-symbol subframes. Complete representations at 19 dB give NeST +1.806 dB PSNR over ARPC, but worse DISTS[1]: $+0.0054\,[0.0024, 0.0086]$. DISTS is an additional perceptual measure in this historical experiment, not a replacement for the four main metrics.

SNR mismatch. With both WHOLE/PARTIAL transmit policies fixed at 10 dB and RX using the true noise variance, all 300 receptions per method on 100 sources are rejected at 8 dB. All succeed at 10 and 12 dB; for each source and method the recovered prefix and reconstruction are identical at those two points. PARTIAL exceeds WHOLE by $0.660\,[0.515, 0.814]$ dB PSNR. These tested points do not locate the rejection threshold or measure the steepness of the 8–10 dB transition.

Processing cost. The following software times are from an RTX PRO 6000 with six CPU cores, batch one, 16 development images, and three measured repetitions after warm-up. They are measured processing times, distinct from the count of length checks. Ada was not timed under this protocol.

Method	SNR (dB)	TX mean (ms)	RX mean (ms)	Total mean (ms)
NeST	4	185.50	358.71	544.21
NeST	10	220.89	398.82	619.71
NeST	19	302.35	472.27	774.62
Latent JSCC	4	16.15	24.50	40.65
Latent JSCC	10	17.97	24.51	42.48
Latent JSCC	19	16.99	24.43	41.42
SwinJSCC	4	24.02	18.59	42.61
SwinJSCC	10	24.81	19.89	44.69
SwinJSCC	19	24.90	18.43	43.32
BPG + LDPC	4	2285.93	18.50	2304.43
BPG + LDPC	10	2269.87	23.15	2293.01
BPG + LDPC	19	2269.24	27.26	2296.50
ARPC + LDPC-QAM	4	716.44	719.05	1435.49
ARPC + LDPC-QAM	10	719.12	801.18	1520.30
ARPC + LDPC-QAM	19	718.19	868.69	1586.89

All underlying means, medians, and 95th percentiles remain in `inputs/timing_summary.csv`.

The existing NeST timing logs also record actual sequential length-fit attempts. Each SNR below has 16 development sources and three measured repetitions (48 calls), with 48 warm-up calls excluded. This is the timing population, not common500. The 95th percentile uses linear interpolation at position $0.95(n-1)$.

SNR	Mean attempts	Median	95th percentile	Range
4	4.7500	5.0	5.0	3–5
10	4.7500	5.0	5.0	3–5
19	4.3125	4.5	5.0	3–5

These are observed candidate-length comparisons after the shared source traversal; their count is neither additional prior forward calls nor the size of the candidate catalogue.

[1] K. Ding, K. Ma, S. Wang, and E. P. Simoncelli, "Image quality assessment: Unifying structure and texture similarity," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 5, pp. 2567–2581, May 2022.

S5.1. Finite Quarter/Eighth timing: incomplete identity validation

Status: the planned 100-source, three-repeat benchmark is not complete. The fixed experiment uses N1024, 10 dB, maximum endpoint $(10, 0)$, 16-QAM rate 2/3, batch one, fixed CPU affinity/threads, and one shared PRO6000. The two grids are interleaved. Its cap is 4 qualification, 6 warmup, and 600 measured TX calls, with a two-hour absolute wall limit including load.

The four real-weight qualification calls matched the historical tokens, all candidate streams and CDF hashes, selected endpoint, and full waveform. The first 88 sources then passed both grids in the first measured repetition. At source position 88 (common500 index 114), fresh tokenizer tokens differed from the historical cache; CDF, stream, endpoint/payload, and waveform comparisons consequently failed. The run stopped immediately. The model weights, source archive, and numerical-route identities had passed their checks. These facts do not establish the root cause of the token discrepancy. No numerical tolerance was used to relabel the event as a pass, and no cached-token substitution or retry was made.

The run performed 187 tokenizer calls, 1870 prior-scale calls and 187 complete TX waveform calls: 4 qualification, 6 warmup, 176 validated measurement calls and one invalid measurement call. The latter remains in the ledger and raw timing CSV; 423 planned TX calls were never attempted. Both workers exited, and no RX, quality metric, noise, training, or bootstrap call was made.

Partial diagnostic only: 88 sources, one repetition per grid. The table excludes the invalid call and all warmup/qualification calls. It must not be combined with the full 100-source quality experiment to infer a Quarter/Eighth quality–cost frontier.

	Quarter (ms)			Eighth (ms)		
Component	Mean	Median	p95	Mean	Median	p95
Tokenizer	19.405	18.324	18.842	19.432	18.356	19.121
Prior/CDF/context	248.160	244.817	255.861	245.131	245.236	250.648
Incremental AC	12.280	12.176	12.822	12.088	12.093	12.466
Endpoint finalization	1.904	1.914	2.022	3.561	3.593	3.809
AC + finalization	14.184	14.073	14.790	15.649	15.682	16.190
Length lookup	0.138	0.131	0.164	0.152	0.145	0.174
Packing/LDPC/QAM	6.751	2.604	59.288	6.272	2.593	30.734
PHY + IPC	7.201	3.055	59.791	6.713	3.022	31.199
Complete TX	289.465	280.975	346.681	287.541	283.283	331.781

Each validated call uses ten prior-scale calls and encodes 680 tokens once. Quarter/Eighth finalize 25/49 streams; mean length checks in this partial subset are 9.534/17.659. The full 100-source quality diagnostic retains its original 9.49/17.58 means. The partial source-paired complete-TX difference (Eighth minus Quarter) is -1.924 ms in mean and $+1.683$ ms in median; these shared-server, incomplete measurements support no speed-ranking claim.

Complete TX runs from resident uint8 RGB to the complete host waveform, including preprocessing/H2D, tokenizer, prior/CDF, incremental AC, non-mutating finalization, length lookup, packing/LDPC/QAM, and persistent CPU-worker IPC. Model loading, file I/O, and durable audit writes are outside the timed interval. CUDA is synchronized at the required boundaries; prior/CDF/context is wall-time attribution, not isolated kernel time. Raw calls, means/medians/p95, per-source differences, resource samples, and original failure/exit receipts are preserved in the V3 `timing/` directory. Earlier resource-gate and moved-checkpoint-path failures occurred before any scientific call and remain separately recorded.

S6. Preserved qualitative cases and source records

The earlier N1024/10 dB development examples, Pastries and Dog, are preserved in S8 alongside the new 4-dB displays. Their full source IDs are:

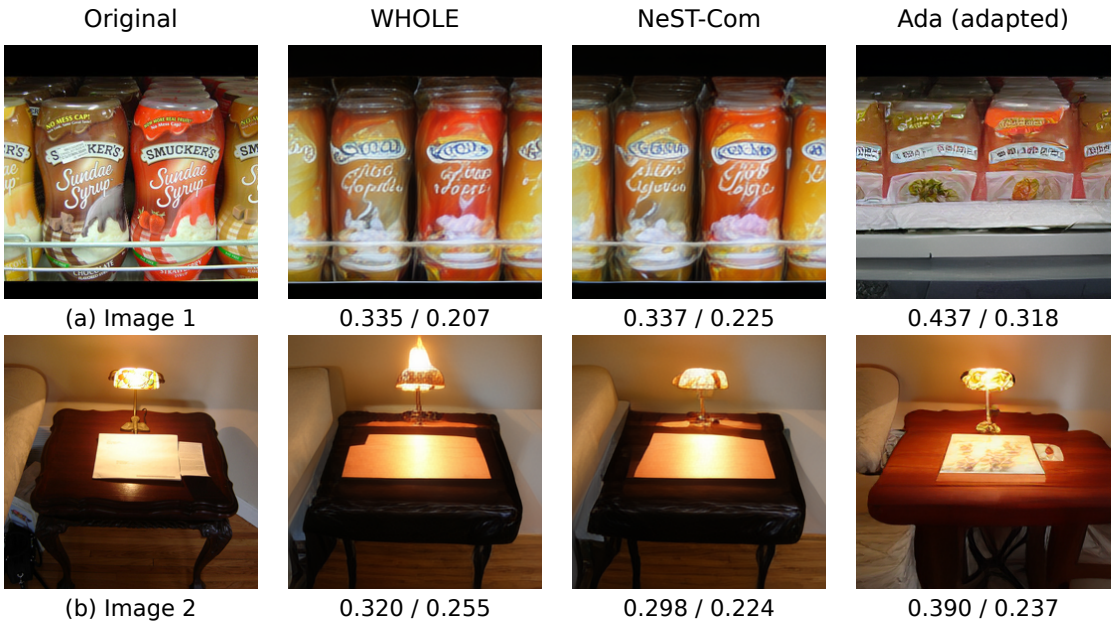
- Pastries: `n07860988/ILSVRC2012_val_00010848_n07860988`
- Dog: `n02105641/ILSVRC2012_val_00018762_n02105641`

They are not newly sampled common500 test examples. NeST/ARPC retain historical noise identity 6201; Ada/Latent JSCC retain 2001, with Ada from its independent development-two-image event. No claim of identical AWGN vectors across these methods is made. Displayed LPIPS-Alex/DreamSim values correspond to each original native-array realization.

The N512 lamp full-field/ROI view is preserved below as a local illustration. The crop is $x \in [80, 180)$, $y \in [0, 100)$ in the original 256-pixel grid and is identical across methods. PSNR/LPIPS values are for each full image, not the crop. It uses N512, 10 dB, original repetition seed 6201; the already displayed lamp was selected editorially, not by a new preregistered or random rule.



The earlier two-case N512 figure is also retained, with full-field images and LPIPS-Alex/DreamSim per displayed realization. Its source-ID selection history is unchanged. Both cases use N512, 10 dB, original repetition seed 6201.



The native-array identifiers, source pixel hashes, wire inputs, display-image hashes, and full-precision metrics are retained in the source map and original selection record. No image content was enhanced or repaired.

Machine-readable evidence

E1 report and complete frame/source/paired data

E2 report and complete frame/source/paired data

100-source statistics identity

N512 formal results and policies

Historical mismatch: index-only bootstrap reuse clarification

Timing data

S7. Versioned N1024 no-fit extension

The historical N1024 experiments used the error-on-no-fit implementation and contained no no-fit event. The independent protocol version `NESTCOM_N1024_PAID_NULL_20261029_V2` adds paid NULL handling without changing those frozen sources or results. It preserves the original 372 profile records, header waveforms, public scrambling convention, and normal source path. Twelve appended IDs cover the original twelve MCS layouts. Missing source evidence remains an error and is not converted to NULL.

The bounded CPU checks use three historical payloads, each through the original and extended catalogues, plus twelve constructed no-fit cases covering all MCS layouts. The ordinary-path full waveforms, profiles, information/codeword hashes, and RX outputs are identical. Each no-fit case sends the full 1024-symbol frame: the paid 68-symbol header and 956-symbol body, including the 13-bit length and CRC16. The receiver recovers the explicit two-zero-bit NULL payload and returns the fixed 0.5 float gray image. Additional pure checks cover full-scale WHOLE/PARTIAL fitting and rejection guards.

Actual checks: 18 full-frame encodes, 18 full-frame receptions and 36 packet decodes; 0 source-model calls, 0 metric calls and 0 neural replays. The process exited successfully after 19.34 seconds.

Three existing image archives are hash-verified and reused because their RX inputs are identical; no new neural output equivalence is claimed from inference. These are finite noiseless-input checks with a 40 dB demapper setting. They establish targeted protocol behavior, not noisy-channel reliability or new image-quality performance. No training or resampling was performed.

The reproducible implementation and input identities are provided in:

- `experiments/nestcom-focused-review-20261029/`
- `results/nestcom_focused_review_20261029_v2/nofit/`

S8. Fixed 4-dB development examples and preserved 10-dB view

The main Fig. 3 uses the same previously selected Pastries (development index 95) and Dog (index 21), at $N = 1024$ and 4 dB. They are not a random or representative common500 sample. The five historical methods use their original first formal timing repetition (`r3`); their ten recovered float arrays match the original recorded hashes exactly. Only the two missing WHOLE and two missing Ada events are new. All four new receptions succeeded. No selection, policy recalibration, training, or population-level statistical test was performed.

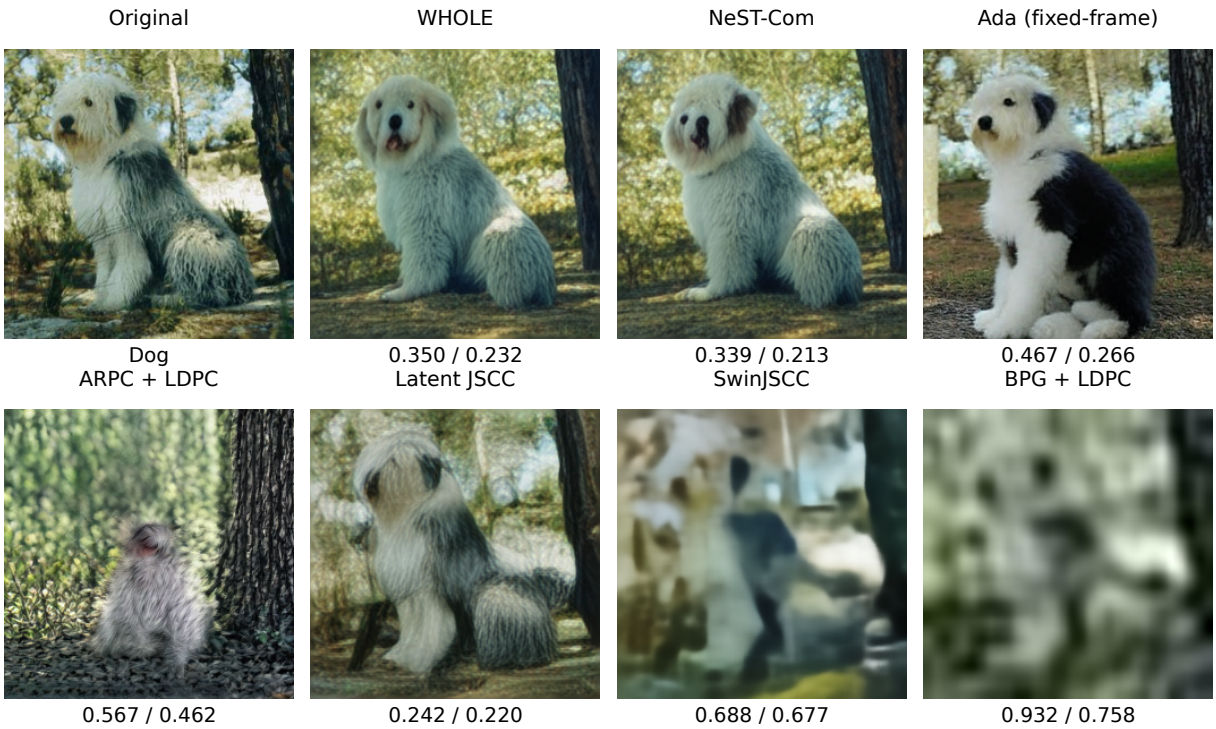
WHOLE uses the frozen target $(7, 0)$ and QPSK rate 2/3, falling back to $(6, 0)$ for both images. It sends 974 source bits per image within the 1245-bit source capacity. Ada reuses the original predicted classes and independently terminated source streams; its paid prefixes contain 103 and 101 tokens for Pastries and Dog. The sampled tail retains CFG=4, $T = 1$, top-$k = 2000$, and top-$p = 1$.

The original method-specific noise identities are retained: NeST/ARPC use 6201 and Latent/Swin/BPG use 2001. The four new visualization events have a separately registered namespace, with WHOLE seed 6201 and Ada seed 2001. Equal seed labels do not assert identical noise vectors across methods. Metrics below are computed from the displayed native float32 reconstructions using the frozen LPIPS-Alex and DreamSim loaders. PNGs use only clipping and rounding to uint8; there is no crop, enhancement, or repair.

For these two images, NeST has lower LPIPS and DreamSim than WHOLE. Both next-scale reconstructions nevertheless change croissant shapes. Latent JSCC has lower LPIPS than NeST in both cases and lower DreamSim for Pastries. These local observations do not replace the common500 results or the 10-dB Ada advantage reported in the main text.

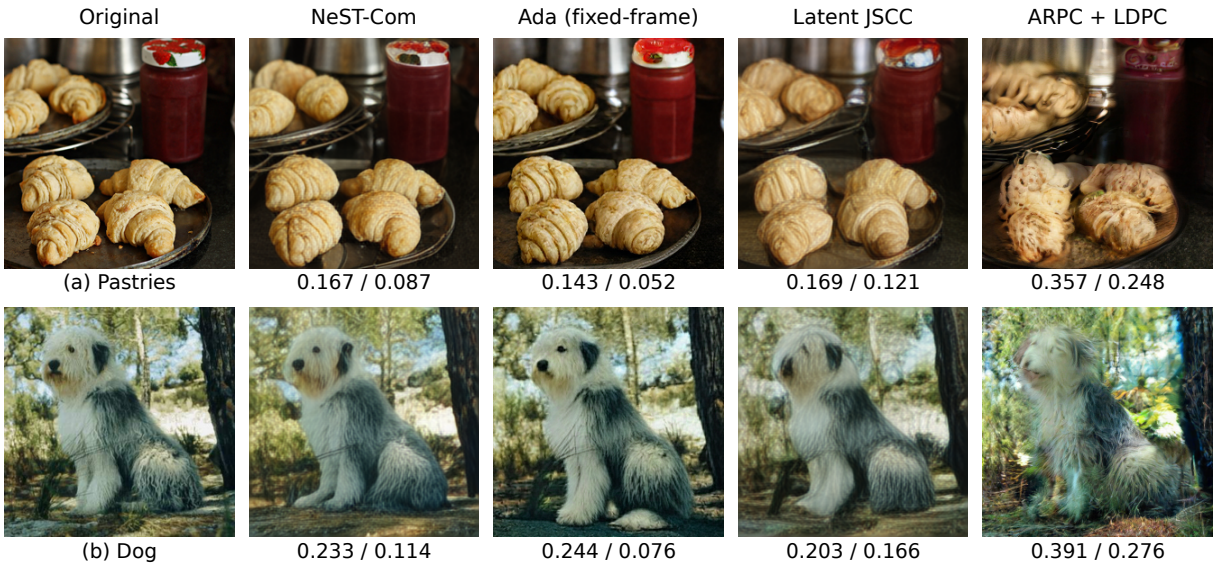


Pastries at 4 dB: original plus all seven systems. Values are LPIPS-Alex↓ / DreamSim↓ for each displayed realization.



Dog at 4 dB: original plus all seven systems. The images are shown at their complete 256×256 field of view. Values are LPIPS-Alex↓ / DreamSim↓ for each displayed realization.

The complete native-array identifiers, paid profiles, source identities, received inputs, noise identities and full-precision metrics are recorded in the accompanying `figure3_metrics.csv` and source mapping. The task adds four physical frames, fourteen receiver renders (ten historical recoveries), fourteen LPIPS pairs and fourteen DreamSim pairs. A WHOLE import-name conflict failed before model loading and was repaired by a separately recorded entry point; the failed receipt is retained.



Preserved original N1024/10-dB Fig. 3, without image or metric changes. Its five columns are Original, NeST-Com, Ada (fixed-frame), Latent JSCC, and ARPC + LDPC. The corresponding metrics remain in `supplemental/N1024_10dB_figure3_metrics.csv`. This display uses the historical development identities described in S6 and is not a matched-noise causal comparison with the new 4-dB visualization events.